

Switch Orchard

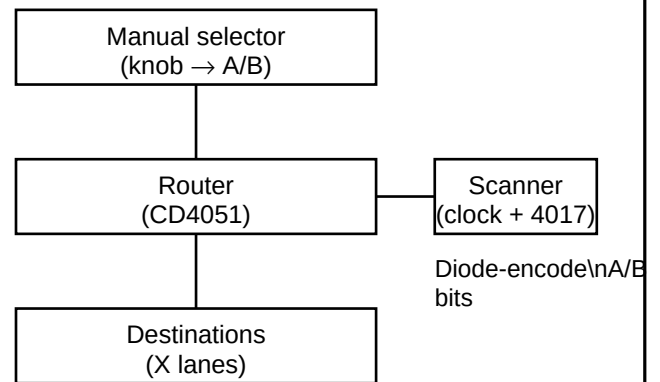
AE Modular BRAEDBOARD • CD4051 routing: choose a path, then let it walk

Core idea: the CD4051 connects the common pin Z to one of X0–X7. Three address lines (A/B/C) choose the lane. Manual mode uses a knob to choose. Scanner mode uses a clock + 4017 to step choices over time.

Build in layers

- 1) Power + decouple the 4051.
- 2) Wire Z + four lanes X0–X3.
- 3) Manual A/B (C=0): knob → thresholds → two bits.
- 4) Verify routing with LEDs or audio.
- 5) Scanner option: clock → 4017 → diode-encode A/B.
- 6) Mode switch: manual vs scanner.

Concept map



Checkpoints

- 4051 powered (VDD=+5, VSS=0, VEE=0).
- INH is LOW (enabled).
- A/B change state (LEDs) as knob turns.
- Z routes to exactly one lane.
- Scanner: A/B cycle 00→01→10→11 with clock.

Patch recipes

- Rotating modulation: X0–X3 = different CVs, Z → destination.
- Audio texture switch: X lanes = audio sources, Z → mix/VCA.
- Demux: Z = one source, X lanes = destinations.
- Rhythm distributor: trigger into Z, scan it across voices.

Build hygiene

Keep A/B/C defined (no floating). Keep INH low. Decouple the 4051. Short address wires prevent ghost switching. Switching audio will click—switching CV is often smoother.